

GenScale: A Benchmark for Relative Object Scale in Image Generation and Editing

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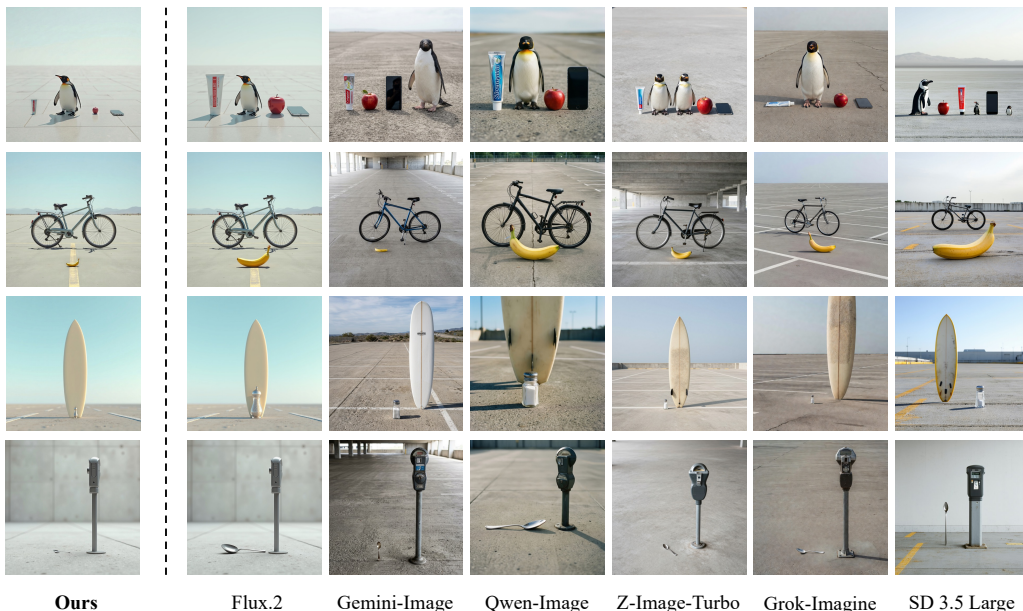


Figure 1: **Relative object scale remains a fundamental challenge in modern image generation.** Even with explicit prompts requiring realistic proportions, state-of-the-art models often fail to render correct real-world size relationships among multiple objects, especially when the scale gap is large. (Zoom in for best viewing.) TODO: add task 2/3 failure cases

Abstract

1 Despite rapid progress in text-to-image generation and image editing, modern generative models still struggle with a basic requirement of physical realism: rendering objects at plausible *relative scale*. Objects may be individually photorealistic and semantically correct, yet their proportions to surrounding objects can violate familiar real-world size relationships. Existing benchmarks usually subsume scale under broader notions of spatial reasoning, prompt following, or physical commonsense, leaving it unclear whether models possess reliable physical-size priors or can repair scale failures after generation. We introduce **GenScale**, a benchmark and evaluation protocol for physically grounded relative object scale in image generation and editing. *GenScale* treats scale as a pairwise relation between objects and evaluates it across three progressively challenging regimes: implicit common-object size priors, explicit metric scale realization, and post-generation scale correction. This pairwise formulation makes scale evaluation intrinsically context-dependent. To reflect these factors, we calibrate an ordinal pairwise evaluator against human annotations and use it as the basis for large-scale model comparison. The resulting evaluation of state-of-the-art open and closed generators and editors exposes a persistent scale-realism gap: models regress object sizes toward typical visual scales

and remain unreliable under explicit metric cues or corrective editing. Beyond diagnosis, we show that *GenScale* feedback can guide *Rescale*, a model-agnostic post-processing framework that improves scale plausibility without modifying the source generator. Together, *GenScale* establishes relative object scale as a targeted, measurable, and actionable dimension of physical realism for image generation and editing.

1 Introduction

Recent text-to-image (T2I) generation has advanced rapidly, with diffusion-based models [41], autoregressive generators [55], and transformer-based systems [8] achieving strong visual fidelity, stylistic diversity, and prompt adherence. As generated images are increasingly used in advertising, product visualization, virtual character creation, and professional content production, evaluation has shifted from global quality and alignment metrics toward capability-focused tests of compositionality [12], physical plausibility [31], and spatial understanding [49]. Yet these broad evaluations do not guarantee correct real-world proportions among objects: an image may look photorealistic and prompt-aligned while still failing at *relative object scale*, placing familiar objects at mutually implausible sizes.

This failure is not a rare corner case of weak models. As illustrated in Fig. 1, strong open and closed generators can preserve object identity and local realism while violating simple size relationships, such as rendering a banana near bicycle scale, a smartphone larger than a penguin, or an explicitly dimensioned product implausibly large relative to a human hand or face. Fig. 2 further shows a consistent directional bias across evaluated generators: small real-world objects are often judged too large, while large real-world objects are often judged too small. We refer to this recurring pattern as *mean regression in generated object scale*, which undermines visually credible generation, product visualization, synthetic data construction, and scale-aware editing.

Existing benchmarks cover several neighboring capabilities but do not isolate this failure mode. They evaluate object presence, counting, and attribute binding [12], compositional prompt following [21], commonsense plausibility [10], physical reasoning [31], and spatial relation control [49]. However, scale is typically treated as one incidental aspect of broader semantic, physical, or spatial reasoning. The finer-grained question we study is *relative object scale*: when several objects appear in a single image, **can a generative model render their size relationships in a way that faithfully reflects the real world?** This question differs from general physical plausibility in granularity: a scene may satisfy coarse commonsense and spatial-layout constraints while still assigning an implausible size ratio to two objects. Because scale is entangled with object identity, placement, perspective, depth, occlusion, and prompt alignment, existing benchmarks make it difficult to diagnose and compare models specifically along this axis.

This gap motivates *GenScale*, a benchmark for evaluating real-world relative object scale in image generation and editing. *GenScale* formulates scale as a pairwise relation between an *anchor* object and a *target* object, grounded in external physical-size metadata rather than prompt text alone. It covers three complementary regimes: implicit common-object size priors, explicit metric product scale relative to human anchors, and post-generation scale correction, where an editor must repair a scale error in an already generated image. Overall, *GenScale* contains 900 image-level entries and 1,643 anchor–target scale relations, enabling dense pairwise evaluation while preserving image-level grouping for model comparison. Tab. 1 summarizes this distinction: prior benchmarks cover compositional, physical, or spatial capabilities, but not the three scale-focused regimes jointly targeted by *GenScale*.

This pairwise formulation also makes evaluation inherently context-dependent: relative scale varies with object identity, physical dimensions, viewpoint, depth, occlusion, and pose. A scale-focused

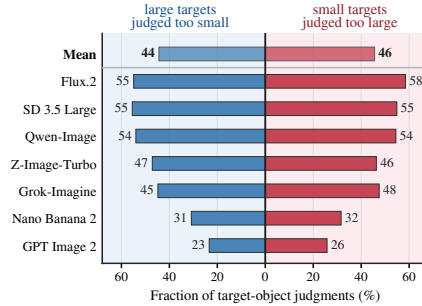


Figure 2: **Mean-regression errors on *GenScale* Task 1.** *GenScale* scores target-object scale from 1 to 5, where 1/2 mean *too small*, 3 means *correct*, and 4/5 mean *too large*. Bars show the fraction of extreme judgments for each generator: large real-world targets judged too small and small real-world targets judged too large. The same error pattern appears across generators, showing that current models systematically compress object scale toward a typical size.

Table 1: **Comparison with representative related benchmarks.** Prior benchmarks cover individual axes such as composition, physical plausibility, or spatial reasoning, whereas *GenScale* explicitly unifies object–object relative scale, human–product metric size, and post-generation scale correction.

Work	Broad Capabilities			Scale-focused Coverage		
	Compos.	Physical	Spatial	Obj.-obj. scale	Human-prod. size	Correction
GenEval [12]	✓	✗	✓	✗	✗	✗
T2I-CompBench++ [21]	✓	✗	✓	✗	✗	✗
GenAI-Bench [26]	✓	✗	✗	✗	✗	✗
PhyBench [31]	✗	✓	✗	✗	✗	✗
GenSpace [49]	✗	✓	✓	✓	✗	✗
<i>GenScale</i>	✓	✓	✓	✓	✓	✓

benchmark, therefore, requires an evaluator that reflects human judgment rather than raw image-space geometry. We therefore develop a human-calibrated ordinal evaluation protocol and calibrate a Gemini-based judge against annotations from nine human raters, achieving strong agreement with human consensus, including Pearson correlation $r = 0.833$. Using this fixed evaluator, *GenScale* reveals persistent relative-scale failures across state-of-the-art generators and editors. Beyond diagnosis, we show that *GenScale* feedback is actionable through **Rescale**, a model-agnostic agentic post-processing pipeline that identifies scale errors, plans localized corrections, and performs geometry-aware resizing and reinsertion without modifying the original generator. Together, these results show that *GenScale* supports both standardized evaluation and downstream correction.

In summary, our contributions are four-fold:

- We identify **relative object scale** as a distinct and underexamined failure mode in modern image generation, and formulate it as a real-world-size-grounded pairwise evaluation problem.
- We introduce **GenScale**, a benchmark for relative object scale across three regimes: implicit common-object size priors, explicit metric product scale in human–product interactions, and post-generation scale correction.
- We develop a **human-calibrated ordinal evaluation protocol** for pairwise scale judgment, and use it to benchmark state-of-the-art generators and editors. The evaluation reveals systematic mean-regression errors, product-magnification bias, and persistent weaknesses in autonomous scale-error correction.
- We demonstrate that *GenScale* feedback is actionable through **Rescale**, a model-agnostic agentic post-processing pipeline that improves relative-scale plausibility across diverse source generators without modifying those generators.

2 Related Work

2.1 Image Generative Models.

Text-to-image generation has progressed rapidly from diffusion-based models [18, 45, 39, 42, 41] and autoregressive generators [55] to large transformer-based diffusion models [36] and rectified-flow generative systems [8]. DALL-E 2 [39], Imagen [42], and latent diffusion models [41] established strong visual fidelity and scalable text-conditioned synthesis, while Parti [55] demonstrated content-rich autoregressive generation from textual prompts. Subsequent transformer-based diffusion models, such as DiT [36], and rectified-flow transformers, such as Stable Diffusion 3 [8], further improved scalability, image quality, and prompt adherence. Recent open and closed systems, including Stable Diffusion 3.5 [46], FLUX.2 [4], Qwen-Image [51, 37], Z-Image [47], Seedream 4.5 [5], GPT Image 2 [34], and Gemini 3 Pro Image [13], extend these advances to higher-fidelity generation, stronger text rendering, reference conditioning, and image editing. This progress motivates evaluating relative scale as a distinct model capability rather than assuming it follows from visual quality.

2.2 Evaluation Benchmarks for Text-to-image Generation.

Early evaluation of image generation relied on distributional image-quality metrics such as FID [17] and IS [43], image-text similarity metrics such as CLIPScore [16], and curated prompt suites such as DrawBench [42] and PartiPrompts [55]. More targeted benchmarks have since evaluated zero-shot compositional generalization [35], object presence, counting, color, position, and attribute binding with GenEval [12], and open-world composition, numeracy, and object relations with T2I-CompBench [20] and T2I-CompBench++ [21]. Recent benchmarks further stress complex

Table 2: **Overview of GenScale.** GenScale evaluates physically grounded relative scale across implicit common-object priors, explicit metric product scale, and scale correction. Image-level entries may contain multiple anchor–target pairs.

Task	Scenario	Images	Pairs	Capability tested
T1	S1: Natural Depth	200	570	Implicit scale priors under perspective
T1	S2: Same Plane	200	573	Implicit scale priors with depth controlled
T2	S3: Human–Product	300	300	Metric product scale with human anchors
T3	S4: Auto-Discovery	100	100	Diagnose scale error and choose resize
T3	S5: Precise Instruction	100	100	Execute exact numeric resize factor
Total		900	1,643	Generation, customization, and correction

prompt following with GenAI-Bench [26], reasoning with R2I-Bench [6], long-prompt detail control with DetailMaster [25], knowledge-intensive factuality with T2I-FactualBench [22], and world-knowledge-informed semantic evaluation with WISE [33]. Automatic evaluation increasingly uses VLM-based protocols such as TIFA [19], VQAScore [29], and GenAI-Bench [26], but recent analyses show that off-the-shelf LMM judges require task-specific validation for nuanced generated-image judgments [56]. In contrast, *GenScale* uses a human-calibrated ordinal protocol for scale-specific pairwise judgment.

2.3 Spatial, Physical, and Commonsense Evaluation.

Relative object scale is also related to spatial reasoning and physical commonsense, which have recently become important evaluation targets for image generation. Commonsense-T2I [10] evaluates everyday commonsense consistency, PhyBench [31] targets physical commonsense errors, and Generate Any Scene [11] evaluates structured object and relation control through scene graphs. Most closely related to our work, GenSpace [49] benchmarks spatially aware image generation through pose, spatial-relation, and metric-measurement tasks, while recent spatial-intelligence benchmarks further emphasize object arrangement and layout consistency [50]. *GenScale* complements these benchmarks with a focused treatment of real-world relative scale, including post-generation scale correction.

3 GenScale: A Benchmark for Physically Grounded Relative Scale

Relative object scale is a basic requirement for physically plausible image generation, yet it is not isolated by existing text-to-image benchmarks. GenScale evaluates whether generators and editors can render realistic real-world proportions by treating scale as a physically grounded pairwise property rather than a generic compositional attribute. Each benchmark entry is associated with structured metadata: object identities, physical reference lengths, anchor–target pairs, expected 3D scale ratios, scenario labels, and, when applicable, product reference images.

GenScale contains three tasks and five scenarios, covering initial generation, product-scale customization, and post-generation scale correction. The full benchmark contains 900 image-level entries and 1,643 pairwise scale relations. Since a single image may contain more than two objects, we use each anchor–target pair as the atomic evaluation unit while retaining image-level grouping for sampling and confidence intervals. Table Tab. 2 summarizes the benchmark taxonomy, and Fig. 3 illustrates the construction pipeline and the input conditions for the five scenarios.

3.1 Physical Scale Knowledge Base

We construct a physical scale knowledge base that maps visual categories to characteristic real-world dimensions. Starting from common visual taxonomies such as COCO [28] and LVIS [15], we retain categories that are visually identifiable, have reasonably stable physical extent, and can be grounded by authoritative external sources. We remove categories with high intra-class size variance, ambiguous semantics, or non-rigid extent, such as *person*, *dog*, *bird*, *tree*, *bag*, *chair*, and *boat*, which are poor scale anchors because their physical sizes vary substantially across instances or subtypes.

The resulting knowledge base contains 100 physical entries: 97 non-human common objects for Task 1 and human anchors, including hand, head/face, foot/leg, and full body, for Task 2. This separation prevents the common-object task from collapsing into human-body scale estimation and makes human–product scale a distinct evaluation regime. Object dimensions are collected from cross-domain sources, including anthropometric surveys for human anchors [14], biological references for animals [32], food and commodity references [48], standardized manufactured-object

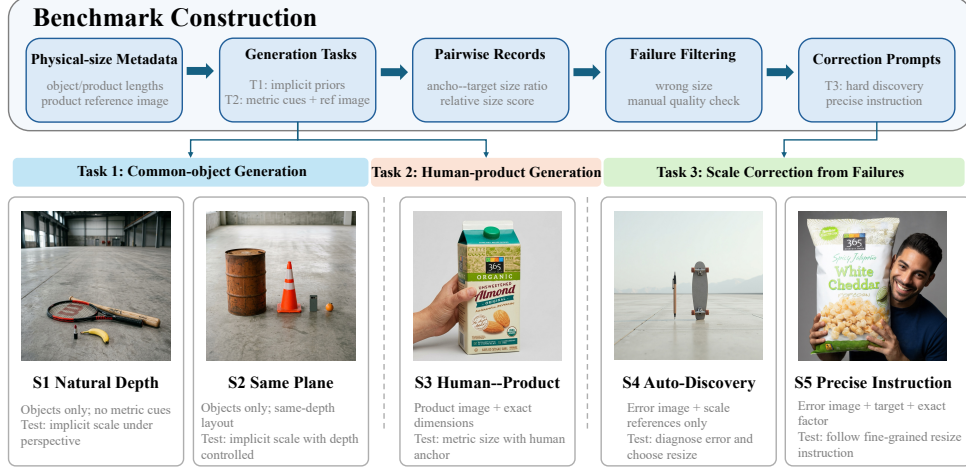


Figure 3: **GenScale benchmark construction.** GenScale is built from physical-size metadata and product references, then instantiated as three tasks covering common-object generation, human-product metric scale, and scale correction. **Task 3** converts non-plausible but editable outputs from **Tasks 1** and **2** into **S4 Auto-Discovery** and **S5 Precise Instruction**, which separately test automatic scale-error diagnosis and exact numeric resize following.

specifications [24, 23, 1], traffic and sport-object specifications [9, 30], and product-level dimensions from Amazon Berkeley Objects (ABO) [7]. For each object, we store a characteristic length, an acceptable physical range when available, and the source used to ground the measurement.

3.2 Benchmark Construction

Task 1: Implicit Common-Object Scale Priors. Task 1 evaluates whether a model has usable commonsense priors about the relative sizes of familiar objects. Each prompt contains two to four non-human objects sampled from the physical scale knowledge base, without numeric dimensions, so the model must infer relative scale from internal object-size knowledge rather than explicit metric cues.

To make each instance diagnostic but visually plausible, we retain object sets whose largest-to-smallest characteristic-length ratio lies in [4, 20]. Ratios below 4 often yield near-equal comparisons that are difficult to judge reliably, while ratios above 20 can produce scenes dominated by layout constraints rather than scale reasoning. Each accepted object set is converted into a photorealistic prompt and evaluated through all valid anchor-target pairs implied by the set.

Task 1 contains two complementary scenarios. In **S1: Natural Depth**, objects may appear in a realistic 3D arrangement with perspective, mild occlusion, and depth ordering, reflecting ordinary generation settings but allowing scale errors to be partially hidden by near-far placement. In **S2: Same Plane**, objects are constrained to lie on approximately the same depth plane, reducing perspective ambiguity and making image-space proportions more directly reflect real-world scale ratios. Together, S1 and S2 separate scale reasoning under natural perspective from scale reasoning when depth ambiguity is minimized.

Task 2: Explicit Metric Scale in Human-Product Interaction. Task 2 targets product-level generation and customization, where a generated product may be visually recognizable but physically implausible relative to the human body. Unlike Task 1, this setting cannot rely on category-level priors alone because visually similar commercial products may have substantially different physical dimensions.

We build Task 2 from ABO products [7] with structured physical dimensions and matched reference images. We retain English product entries with valid length-width-height metadata, convert all measurements to centimeters, and use the maximum axis extent as the characteristic product length. Each instance provides the product reference image and explicit metric size cues in the prompt, requiring the model to preserve product identity while rendering a plausible scale relative to a human anchor.

The human anchor is chosen adaptively from the product size: products shorter than 30 cm are paired with a hand, products between 30 and 60 cm with a head/face anchor, products between 60 and 100 cm with a foot/leg anchor, and products larger than 100 cm with a full-body anchor. Prompts

require natural, use-consistent interaction and catalog-style visibility rather than arbitrary object juxtaposition, testing whether models can translate metric dimensions into plausible human-object proportions.

Task 3: Scale Correction from Failed Generations. Task 3 evaluates scale-aware image editing rather than initial generation. It separates two capabilities that are often conflated: diagnosing an implausible relative-scale relation and choosing a local size correction, versus executing a fully specified fine-grained resize instruction. We build Task 3 from non-plausible but editable outputs from Tasks 1 and 2, using FLUX.2 [3] generations for common-object scenes and Qwen-Image [51] generations for human-product scenes. To make correction well-defined, we keep only pairwise source cases: two-object Task 1 images and one product-human relation in Task 2.

For each source image, we use the GenScale pairwise evaluation record to select a relation with ordinal score $\neq 3$. The same record provides the rendered object-size ratio and target physical ratio, from which we derive a correction plan specifying the fixed reference object, editable object, resize direction, and multiplicative scale factor. We remove images with missing or unidentifiable objects, duplicated or merged objects, severe artifacts, heavy occlusion or blur, and extreme correction factors that make localized editing ill-defined. This filtering is independent of downstream editing success and only ensures that each source image contains a scorable and editable scale error.

Each of the 100 retained source images is expanded into two editing scenarios. In **S4: Hard Auto-Discovery**, the model receives the erroneous image and scale references needed to judge the scene, but not the resize direction or scale factor. For Task 1 sources, it must infer which object size should change; for Task 2 sources, the product is the editable candidate and the human body part serves as the fixed anchor. In **S5: Precise Scale Instruction**, the prompt directly gives the editable object, reference object, resize direction, and exact scale factor, isolating fine-grained numeric scale-instruction following from visual error diagnosis.

3.3 Human-Calibrated Scale Evaluation

GenScale evaluates relative scale at the object-pair level, where a purely geometric pixel-ratio metric is insufficient: the same image-space ratio can be correct or incorrect depending on object identity, depth ordering, foreshortening, occlusion, and camera perspective. We therefore use a human-calibrated ordinal protocol. Each generated image is decomposed into anchor-target pairs from the GenScale metadata; the anchor is treated as the reference, and the evaluator judges whether the target is rendered at a plausible real-world size relative to it.

Pairwise ordinal rubric. For each anchor-target pair, evaluators are shown the image, object names, and reference physical lengths. They assign a five-point ordinal score: 1/2 indicate that the target is severely/slightly undersized, 3 indicates a physically plausible scale, and 4/5 indicate that it is slightly/severely oversized. Operationally, score 3 corresponds to an estimated scale error within approximately $\pm 20\%$, scores 2/4 to errors between 20% and 60%, and scores 1/5 to errors larger than 60%.

Evaluators are instructed to account for perspective, occlusion, partial visibility, and foreshortening. A pair is marked invalid only when scale judgment is not possible, e.g., because an object is missing, merged, duplicated ambiguously, or too degraded to identify.

Human consensus. We collect annotations from nine human raters on a calibration split sampled from Tasks 1 and 2, covering natural-depth common-object scenes, same-plane common-object scenes, and human-product scenes. After aligning completed annotations and retaining visible, scorable pairs, the calibration set contains 285 images and 527 anchor-target pairs. For each pair, we define human consensus as the modal score, with ties broken by the median. As shown in Table Tab. 3, individual raters agree strongly with the aggregate consensus, and the leave-one-rater-out comparison remains stable, indicating that the consensus is not dominated by any single annotator.

Automatic judge calibration. Because exhaustive human evaluation is impractical for large-scale model comparison, we calibrate a Gemini-based VLM judge against the human consensus. The judge receives the same information as human raters: the generated image, anchor and target names,

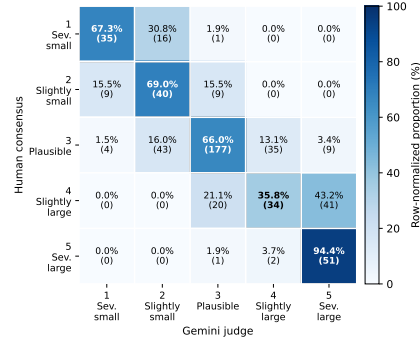


Figure 4: **Disagreement structure of the calibrated Gemini judge.** The row-normalized confusion matrix shows that Gemini-human disagreements are concentrated near the diagonal, with only 2.85% of pairs differing by two or more score levels and no coarse under-to-over or over-to-under sign-flip errors.

Table 3: **Human reliability and Gemini-human alignment.** Exact and ≤ 1 denote agreement on the five-point ordinal scale. MAE is the mean absolute ordinal deviation. r is Pearson correlation on 1–5 scores. QWK denotes quadratic-weighted kappa. Human rows report mean $\pm$ std across nine annotators.

Evaluator / reference	Split	Exact $\uparrow$	≤ 1 $\uparrow$	MAE $\downarrow$	r $\uparrow$	QWK $\uparrow$
Human vs. aggregate consensus	T1+T2	65.1 $\pm$ 7.0	94.0 $\pm$ 3.2	0.413 $\pm$ 0.099	0.764 $\pm$ 0.076	0.754 $\pm$ 0.077
Human vs. LOO consensus	T1+T2	58.0 $\pm$ 6.6	93.3 $\pm$ 3.0	0.491 $\pm$ 0.093	0.719 $\pm$ 0.069	0.706 $\pm$ 0.070
Gemini vs. human consensus	T1+T2	63.95	97.15	0.389	0.833	0.823
Gemini vs. human consensus	T1	61.83	96.49	0.417	0.847	0.837
Gemini vs. human consensus	T2	73.00	100.00	0.270	0.470	0.468

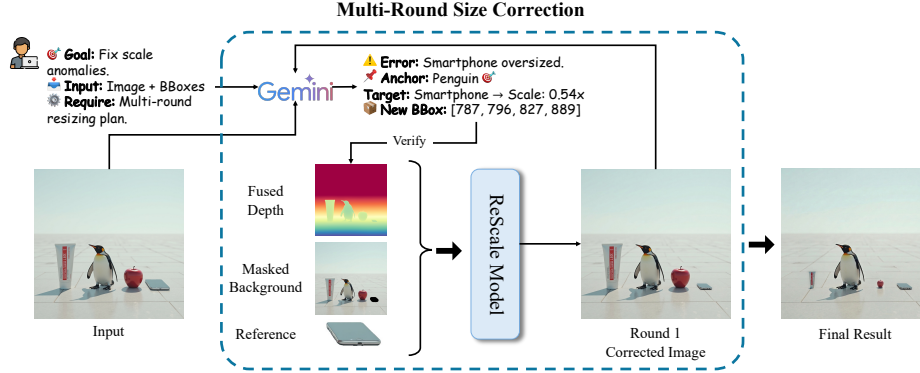


Figure 5: **Agentic multi-round inference in ReScale.** Given a generated image and object/scale metadata, the agent diagnoses a scale anomaly, selects the target and anchor, predicts a resize factor and contact-preserving anchor point, and executes localized correction through extraction, background completion, mask/depth adjustment, and modular insertion. The edited result is verified for additional correction rounds when needed.

physical reference lengths, expected 3D length ratio, and the five-point rubric. The prompt is scenario-specific: Task 1 distinguishes natural-depth from same-plane scenes, while Task 2 treats human body parts as approximate physical anchors and accounts for product packaging, foreshortening, cropped body parts, and catalog-style composition. We query the judge five times per pair and aggregate by majority vote, using the median when no unique mode exists; extreme scores are accepted only when consistently supported across repeated samples.

Before pairwise scoring, we apply a scene-level validity prefilter and skip images only when reliable scale judgment is impossible, such as duplicated target categories, major unnamed objects, severe artifacts, or unclear object boundaries. Ordinary scale errors are still scored. Table Tab. 3 shows that the calibrated judge reaches inter-human-level alignment with the consensus, with especially strong within-one agreement and QWK on the full calibration split. Figure Fig. 4 further shows that remaining disagreements are ordinally local rather than directionally reversed. The lower correlation metrics on Task 2 mainly reflect label concentration near the plausible-scale score rather than systematic judge failure. We therefore use the calibrated Gemini judge as the default evaluator for large-scale GenScale comparisons, while reporting scored-pair coverage because invalid or unscorable generations are excluded by the same prefilter across systems.

4 ReScale: Agentic Relative-Scale Correction

The GenScale formulation makes relative-scale errors actionable beyond evaluation. To fix images with incorrect relative scales, we introduce **ReScale**, a model-agnostic post-processing pipeline that repairs scale inconsistencies in generated images without modifying the source generator. Given an image and object/physical-size references from the prompt or benchmark metadata, ReScale edits only the implausibly scaled object while preserving identity, layout, lighting, and background. At inference time, a multimodal agent localizes candidate objects, selects an editable target o_t and fixed anchor o_a , refines the target box b_t , and estimates a conservative resize factor α by comparing the apparent target–anchor ratio with the plausible real-world ratio under perspective, depth, visibility, and contact constraints. The resulting plan $\pi = \{o_t, o_a, b_t, \alpha, p\}$ specifies the target, anchor, refined box, resize factor, and contact-preserving anchor point p ; if no visible scale inconsistency is detected, the agent returns no edit.

Table 4: **Task 1: common-object relative scale.** Scale error is mean $|s - 3|$. Plausible denotes score 3; severe denotes scores 1 or 5. MR error measures small targets judged too large or large targets judged too small.

Model	Valid img. / pairs	Scale error ↓	Plausible (%) ↑	Too small (%) ↓	Too large (%) ↓	Severe (%) ↓	MR error (%) ↓
Gemini Image Preview [13]	395 / 1121	0.598	56.5	21.1	22.5	16.2	31.8
GPT-Image-2 [34]	399 / 1136	0.706	51.9	21.7	26.3	22.5	24.1
Z-Image-Turbo [47]	374 / 1004	0.755	46.1	27.6	26.3	21.6	45.3
Grok Image [52]	398 / 1136	0.790	46.7	25.5	27.8	25.7	44.2
Qwen-Image-2.5 [37]	395 / 1105	0.905	40.0	29.2	30.8	30.5	52.9
FLUX.2 [4]	398 / 1070	1.025	33.9	33.2	32.9	36.4	55.1
SD3.5-Large [46]	346 / 949	1.065	31.1	34.4	34.6	37.6	57.8

Table 5: **Task 2: human-anchored product scale.** Each prompt contains one product–human scale relation. Too small and too large refer to the product relative to the human anchor.

Model	Valid img. / pairs	Scale error ↓	Plausible (%) ↑	Too small (%) ↓	Too large (%) ↓	Severe (%) ↓
GPT-Image-2 [34]	294 / 294	0.231	78.9	3.4	17.7	2.0
Gemini Image Preview [13]	295 / 295	0.268	75.6	4.4	20.0	2.4
Seedream v4.5 [5]	295 / 295	0.302	74.9	8.1	16.9	5.1
Qwen-Image-Edit-2.5 [38]	297 / 297	0.327	71.0	4.4	24.6	3.7
FLUX.1 Kontext-dev [2]	291 / 291	0.423	63.2	15.1	21.6	5.5
SD3.5-Large + IP-Adapter [46, 54]	270 / 270	0.600	52.6	7.8	39.6	12.6

The planned edit is executed by a modular local insertion pipeline. We segment the target with SAM 2 [40], extract it as an identity reference, remove the original instance to obtain a completed background, construct an enlarged edit mask around the resized box, and estimate a monocular depth map with Depth Anything V2 [53]. These steps produce backend-agnostic conditions, including the reference crop, completed background, resized mask, text instruction, and, when supported, fused depth. Our default implementation uses Insert Anything [44], but the backend can be replaced by other insertion or inpainting editors; Appendix ?? analyzes this modularity. After each edit, the agent verifies whether an obvious scale error remains and applies another conservative round only when needed. Thus, ReScale uses GenScale’s physical-size structure as a correction interface, while the GenScale judge is reserved for evaluation in Sec. 5.2, allowing us to test whether relative-scale knowledge supports generator-agnostic correction rather than only post-hoc ranking.

5 Evaluation Results

5.1 Benchmarking State-of-the-Art Models on GenScale

We evaluate representative state-of-the-art generators and editors on the three GenScale tasks using the calibrated judge from Section Sec. 3.3. Task 1 benchmarks text-to-image generation, while Tasks 2–3 evaluate image-conditioned generation and editing. Unless otherwise stated, all metrics are computed at the pair level after the visual-quality filter in Section Sec. 3.3; missing, duplicated, ambiguous, or severely flawed target/reference objects are excluded before scale scoring. Thus, valid counts measure scorable coverage, while scale metrics measure performance conditional on scorable outputs. For a scale score $s \in \{1, \dots, 5\}$, where 3 denotes plausible scale, we define *Scale error* as mean $|s - 3|$ and report plausible, severe, and directional error rates.

Task 1: common-object relative scale. Task 1 tests implicit common-object size priors without explicit metric cues. Table Tab. 4 shows that relative scale remains far from solved even for strong generators. The main failure pattern is not random: most models exhibit mean-regression behavior, enlarging physically small targets and shrinking physically large targets. We quantify this with the mean-regression error rate, $\frac{1}{2} [P(s \in \{4, 5\} \mid \ell_t < \ell_a) + P(s \in \{1, 2\} \mid \ell_t > \ell_a)]$, where ℓ_t and ℓ_a are the physical lengths of the target and anchor objects. Closed-source models generally perform better, but none eliminates this systematic bias; weaker models also suffer from lower scorable coverage, making their scale scores optimistic relative to all attempted generations.

Task 2: human-anchored product scale. Task 2 evaluates whether models can translate explicit product dimensions into plausible human–object proportions. As shown in Table Tab. 5, this setting is easier than common-object scale reasoning but still not saturated. Errors are directionally asymmetric: models more often enlarge the product than shrink it, suggesting a product-magnification bias in reference-conditioned generation. This is particularly problematic for product visualization, where preserving identity without preserving physical size can still yield misleading outputs.

Task 3: scale-error correction. Task 3 evaluates whether editors can repair relative-scale failures after generation. Table Tab. 6 shows that general-purpose editors can improve failed images, but scale correction remains difficult. The consistent gap between S4 and S5 indicates that executing an

Table 6: **Task 3: scale-error correction.** S4 requires automatic error discovery; S5 provides the target object, reference object, correction direction, and scale factor. Gain is the matched-pair reduction in scale error relative to the before-edit image.

Model	Scored pairs	Scale error ↓	Plausible (%) ↑	S4 error ↓	S5 error ↓	Gain ↑	Better / Worse
Before edit	196	1.265	20.4	1.265	1.265	–	–
GPT-Image-2 [34]	195	0.682	53.3	0.959	0.408	+0.601	91 / 9
Gemini Image Preview [13]	197	0.980	32.5	1.030	0.929	+0.289	55 / 11
FLUX.1 Kontext-dev [2]	193	1.130	28.5	1.293	0.957	+0.126	35 / 16
Qwen-Image-Edit-2511 [38]	190	1.132	28.4	1.253	1.000	+0.132	32 / 9
Seedream v4.5 [5]	196	1.138	31.6	1.170	1.104	+0.150	43 / 17
SD3.5-Large + IP-Adapter [46, 54]	74 [†]	1.230	25.7	0.933 [†]	1.305	-0.167	10 / 19

[†]SD3.5-Large + IP-Adapter has substantially lower scored coverage, especially in S4; its per-scenario results should therefore be interpreted cautiously.

Table 7: **Task 1 correction on common-object generations.** Metrics are computed on matched scorable pairs before and after ReScale correction. Short names are used for compactness: Gemini, GPT, Z-Image, Grok, Qwen, FLUX, and SD3.5 denote the corresponding models evaluated in Table Tab. 4.

Metric	Gemini	GPT	Z-Image	Grok	Qwen	FLUX	SD3.5
Error before ↓	0.592	0.692	0.759	0.785	0.916	1.025	1.041
Error after ↓	0.383	0.445	0.431	0.450	0.423	0.529	0.635
Gain ↑	+0.208	+0.246	+0.328	+0.335	+0.493	+0.495	+0.406

explicit resize instruction is much easier than autonomously discovering the scale error and choosing an appropriate correction. Thus, current editors are better at following localized resize commands than at diagnosing physically implausible scale relations.

Overall, GenScale exposes three complementary failure modes: mean-regression in common-object scale, product magnification in human-product generation, and weak autonomous diagnosis in scale correction. These results show that relative scale is not merely a byproduct of general visual fidelity or prompt adherence, but a distinct capability that current generators and editors do not yet reliably solve.

5.2 ReScale Correction Results

We evaluate ReScale by applying it to generated images and scoring matched before/after pairs with the fixed GenScale judge. Tables 7 and 8 summarize correction results on common-object generation, human-product generation, and the dedicated scale-correction benchmark.

On Task 1, ReScale reduces scale error for every source generator, with larger gains on models that previously exhibit stronger mean-regression behavior. This suggests that many common-object scale errors are not irreversible generation failures, but can be corrected through localized resizing and reinsertion. On Task 2, ReScale also improves all image-conditioned source models, showing that explicit metric product-scale errors are locally correctable even when the original model preserves product identity but misestimates physical size. On Task 3, ReScale outperforms general-purpose editors on the same correction inputs, indicating that explicit scale diagnosis and structured local editing are better suited to this task than generic image editing. Overall, these results show that GenScale’s physical-size structure is actionable, while the remaining errors indicate that autonomous scale repair remains nontrivial.

Identity and visual-quality preservation. Because scale correction should not trade geometric plausibility for image degradation, we additionally evaluate identity preservation and visual quality between the original and corrected images. We report reference-consistency metrics and no-reference image-quality metrics in Table Tab. 9; final values will be inserted after the full quality-evaluation run finishes.

Overall, these results show that GenScale’s physical-size structure is actionable: an automatic, model-agnostic correction agent can substantially improve relative-scale plausibility across diverse source generators without retraining those generators. The remaining gap to perfect correction, especially on autonomous Task 3 cases, indicates that scale repair remains nontrivial; nevertheless, the consistent matched-pair gains demonstrate that GenScale can function not only as an evaluation benchmark but also as a diagnostic foundation for downstream correction systems.

Table 8: **Correction results on edit-oriented tasks.** Task 2 reports ReScale correction on human-product generations from each source model; Task 3 compares ReScale with general-purpose editors on the correction benchmark. Metrics are computed on matched scorable pairs. Short names are used for compactness: Gemini, GPT, Seedream, Qwen, FLUX, and SD3.5 denote the corresponding edit-capable models evaluated in Table Tab. 5 and Table Tab. 6.

Task	Metric	Gemini	GPT	Seedream	Qwen	FLUX	SD3.5	ReScale
Task 2	Error before ↓	0.261	0.231	0.295	0.306	0.406	0.565	–
	Error after ↓	0.086	0.090	0.168	0.144	0.228	0.256	–
	Gain ↑	+0.175	+0.141	+0.126	+0.162	+0.178	+0.309	–
Task 3	Error before ↓	1.263	1.275	1.280	1.259	1.247	1.042[†]	1.258
	Error after ↓	0.974	0.674	1.130	1.127	1.121	1.208	0.548
	Gain ↑	+0.289	+0.601	+0.150	+0.132	+0.126	-0.167	+0.710

[†]SD3.5 has substantially lower matched coverage on Task 3, so its before-edit error is not directly comparable to other editors.

Table 9: **Identity preservation and visual quality after correction.** CLIP-I and DINO measure visual/identity consistency with the pre-edit images. SSIM and SSIM-HF measure image-level and high-frequency preservation. LAION-Aes and Q-Align-IQ measure no-reference visual quality.

Setting	CLIP-I ↑	DINO ↑	SSIM ↑	SSIM-HF ↑	LAION-Aes (before / after) ↑	Q-Align-IQ (before / after) ↑
Task 1 correction	94.8	90.2	88.8	92.4	5.83 / 5.73	4.74 / 4.67
Task 2 correction	92.4	84.5	73.5	82.1	5.00 / 4.96	4.88 / 4.88
Task 3 correction	95.6	88.9	89.3	92.9	5.62 / 5.55	4.76 / 4.67

5.3 Limitations and Future Work

GenScale focuses on relative object scale rather than all aspects of physical or spatial realism, and its physical knowledge base is limited to visually identifiable categories with relatively stable canonical dimensions. Highly deformable, fine-grained, or strongly context-dependent objects remain outside the current scope, and scale judgments may require further validation under broader viewpoints, occlusions, domains, and future model generations. ReScale further assumes that the relevant objects are visible and locally editable; future work could incorporate physical-size priors, metric supervision, or scale-aware preference optimization during training so that generators learn to avoid scale errors rather than only correcting them post hoc.

6 Conclusion

In conclusion, we introduced *GenScale*, a benchmark and evaluation protocol for physically grounded relative object scale in image generation and editing. By isolating scale as a pairwise object relation, *GenScale* reveals that even strong contemporary models often fail to preserve realistic real-world proportions, despite producing visually appealing and semantically correct images. Our results further show that scale-aware feedback can support model-agnostic post-generation correction, suggesting that relative scale is not only measurable but also actionable. We hope that *GenScale* will encourage future work to treat object scale as a core dimension of physical realism, motivating generation models with stronger metric understanding and more reliable real-world size priors.

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Table 10: **Detailed composition of GenScale.** GenScale contains three evaluation tasks and five scenarios. Task 1 evaluates implicit common-object size priors, Task 2 evaluates explicit metric scale realization in product–human interactions, and Task 3 evaluates scale correction on previously failed generations.

Task	Scenario	#Entries	#Pairwise ratios	Object-count distribution	Primary capability tested
T1	S1: Natural Depth	200	570	2:93, 3:55, 4:52	Relative scale under natural perspective and occlusion
T1	S2: Same Plane	200	573	2:81, 3:74, 4:45	Relative scale with perspective largely controlled
T2	S3: Human–Product Anchor	300	300	2:300	Product scale realization with explicit metric cues and reference image
T3	S4: Hard Auto-Discovery	100	100	2:100	Error diagnosis and corrective resizing
T3	S5: Precise Scale Instruction	100	100	2:100	Exact scale-instruction following
Total	–	900	1643	–	–

Table 11: **Coverage of the 100-entry physical scale knowledge base.** Objects are grouped by semantic domain and dimensional source type. The knowledge base contains 42 COCO-derived and 58 LVIS-derived categories, covering both everyday objects and physically standardized objects.

Group	#Obj.	%	Example categories	Dimensional grounding
Human body-part anchors	3	3.0	hand, face, foot	ANSUR II anthropometric measurements
Animals	13	13.0	elephant, giraffe, horse, cat, turtle	Animal Diversity Web and species-level references
Sports / recreation	14	14.0	baseball, basketball, tennis racket, surfboard	Official sport rules and standardized equipment dimensions
Electronics / appliances / media	10	10.0	keyboard, laptop, TV, microwave, CD	ANSI, ISO, and typical product specifications
Office / personal small objects	16	16.0	battery, credit card, passport, pencil, toothbrush	ISO, ANSI, and common product standards
Tableware / containers	11	11.0	bottle, wine glass, mug, fork, bowl, can	Standard tableware and container dimensions
Food items	7	7.0	apple, orange, banana, carrot, egg	USDA and common food-size standards
Vehicles / traffic / public infrastructure	12	12.0	bicycle, car, bus, stop sign, fire hydrant	Vehicle, traffic, and infrastructure specifications
Household / tools / accessories / instruments	14	14.0	umbrella, suitcase, hammer, watch, guitar	Common product and instrument dimensions
Total	100	100.0	–	–

Supplementary Material

Overview

This appendix is organized as follows:

?? gives more implementation details of *Rescale*. Sec 4.1

.1 Additional Benchmark Construction Statistics

Table 11 summarizes the semantic and source coverage of the physical scale knowledge base used by GenScale. The goal of the knowledge base is not to maximize taxonomic breadth, but to retain categories whose physical dimensions are sufficiently stable and externally grounded for relative-scale evaluation. We therefore include objects with standardized or well-documented dimensions, such as sports equipment, tableware, appliances, traffic objects, and common food items, while avoiding categories with large intra-class variation or ambiguous visual extent. The resulting 100-entry knowledge base contains 42 COCO-derived and 58 LVIS-derived categories. It provides the common-object scale priors used in Task 1, while human body-part anchors provide physically interpretable references for product-scale prompts in Task 2.

Table 12 reports the composition of Task 1. Unlike binary-only relative-scale benchmarks, GenScale includes prompts with two, three, and four objects. For an n -object prompt, we evaluate all unordered object pairs, yielding $\binom{n}{2}$ pairwise scale ratios. This design increases relation density while keeping the number of generated images manageable. Task 1 contains 400 prompts and 1,143 evaluated pairwise relations in total, with 570 relations from S1 Natural Depth and 573 from S2 Same Plane.

Table 12: **Task 1 object-count distribution.** Each n -object prompt induces all $\binom{n}{2}$ unordered pairwise scale relations. Thus, three- and four-object prompts substantially increase the number of evaluated relations without increasing the number of generated images.

Scenario	2 objects	3 objects	4 objects	Total
S1 Natural Depth	93 images / 93 pairs	55 images / 165 pairs	52 images / 312 pairs	200 images / 570 pairs
S2 Same Plane	81 images / 81 pairs	74 images / 222 pairs	45 images / 270 pairs	200 images / 573 pairs
Task 1 total	174 images / 174 pairs	129 images / 387 pairs	97 images / 582 pairs	400 images / 1,143 pairs

Table 13: **Human-anchor distribution in Task 2.** Each Task 2 prompt contains one product and one human anchor, resulting in one evaluated product-to-anchor scale relation per image. Product length is computed from the stored `product_scale.typical_len_cm` field. The mean target/anchor ratio is computed from the ground-truth pairwise ratio stored in the benchmark JSON.

Human anchor	Entries	%	Median product length	Product length range	Mean target/anchor ratio
Human hand	151	50.3	18.29 cm	5.08–30.23 cm	0.94
Human head / face	77	25.7	39.62 cm	30.00–58.42 cm	1.72
Human foot / leg	31	10.3	70.61 cm	31.75–96.52 cm	2.81
Full human body	41	13.7	150.00 cm	100.00–243.84 cm	0.95
Total	300	100.0	–	–	–

529 The multi-object prompts are important because relative scale errors often compound when a model
530 must simultaneously maintain several object-size relationships in a coherent scene.

531 Table ?? summarizes the human-anchor distribution in Task 2. Task 2 evaluates product-scale realiza-
532 tion rather than common-object scale priors, so each prompt pairs one product with a human anchor
533 that makes the product dimension visually interpretable. The anchor distribution is intentionally
534 size-conditioned rather than uniform: small products are most often paired with hands, medium
535 products with the head or face, larger floor- or leg-scale products with the foot or leg, and very large
536 products with the full human body. This design reflects realistic product photography and advertising
537 settings, where the chosen human reference should make the target product scale visible without
538 dominating the composition. Each Task 2 entry contributes exactly one product-to-anchor relation,
539 yielding 300 evaluated pairwise relations.

540 For each object pair (i, j) , we define the target relative scale as $r_{ij} = l_i/l_j$, where l_i denotes the
541 characteristic physical length of object i . To account for natural intra-class variation and measurement
542 uncertainty, each pair is also assigned an acceptable interval $[l_i^{\min}/l_j^{\max}, l_i^{\max}/l_j^{\min}]$. For Task 2, the
543 product length and interval are taken from ABO metadata, while the human anchor is assigned a
544 fixed canonical length with a small tolerance.

545 For Task 2, we use ABO as the sole source of commercial products because it provides structured
546 product-level dimensions and product images. We keep products with English metadata, valid
547 local reference images, and complete length/width/height metadata, converting all dimensions to
548 centimeters. We use the maximum axis as the characteristic length and retain products whose
549 characteristic length lies within a practical range for human-scale visualization. The final benchmark
550 samples 300 product–human prompts from this cleaned candidate pool.

551 Task 3 intentionally evaluates failure-case correction rather than average-case editing. This design
552 reflects a practical use case: generated images often contain otherwise high-quality content but fail
553 due to a single implausible object-scale relationship. We therefore select source images where the
554 main objects are recognizable and the scene is editable, while excluding cases where scale correction
555 would be confounded by missing objects, duplicate categories, or severe synthesis artifacts.

556 **Human annotation interface.** We provide the screenshot and full instruction text of the GenScale
557 human evaluation GUI in Appendix ?. Annotators were shown the generated image, anchor and
558 target names, reference physical lengths, and three quality-control checkboxes: whether both objects
559 are visible, whether duplicate objects interfere with scoring, and whether unnamed or merged objects
560 make the pair ambiguous.

Table 14: **Prompt and ground-truth construction.** GenScale separates implicit common-object scale priors from explicit metric scale control.

Task	Input modality	Scale information in prompt	Ground-truth ratio	Evaluation target
T1: Common objects	Text-only generation	No object-specific numeric size is provided. Prompts only request physically accurate real-world proportions.	For objects i, j , $r_{ij} = l_i/l_j$, with acceptable interval $[l_i^{\min}/l_j^{\max}, l_i^{\max}/l_j^{\min}]$.	Whether the model’s implicit real-world object-size prior is accurate.
T2: Human-product	Product reference image + text prompt	Product dimensions from ABO metadata are explicitly included; the human anchor has a fixed canonical size.	Product-to-anchor ratio computed from ABO characteristic length and anchor length, with tolerance intervals.	Whether the model can realize explicit metric scale while preserving product identity and natural human interaction.
T3: Scale correction	Erroneous generated image + edit instruction	S4 provides reference lengths and asks the model to decide whether/how to resize; S5 gives the target resize direction and scale factor.	Same pairwise ratio as the corresponding T1/T2 source case.	Whether an editing model can diagnose and correct scale errors without changing identity, pose, or scene composition.

Table 15: **Task 1 sampling protocol.** Task 1 samples 2–4 common objects and filters combinations by physical disparity.

Design choice	Implementation
Object pool	97 non-human common objects from the physical knowledge base. Human body parts are excluded from Task 1.
Number of objects	Each prompt contains 2–4 objects. In the final benchmark, T1 contains 174 two-object prompts, 129 three-object prompts, and 97 four-object prompts.
Disparity range	Let $D = \max_i l_i / \min_i l_i$ among sampled objects. We keep combinations with $4 \leq D \leq 20$.
Motivation	$D < 4$ yields visually subtle scale differences; $D > 20$ often produces scenes that are physically difficult to compose or visually dominated by one object.
Scenarios	S1 permits natural depth and perspective; S2 constrains objects to a similar depth plane to reduce perspective-based shortcuts.
Scale supervision	No numeric object size is shown to the model; the task evaluates implicit common-object size priors.

561 **Automatic judge prompt.** Appendix ?? gives the full Gemini prompt used for Task 1 and Task 2.
562 The prompt was fixed before evaluating the full benchmark. We use five repeated judge calls per pair
563 and aggregate scores by majority vote, with median tie-breaking.

564 **Calibration diagnostics.** Appendix ?? reports full confusion matrices, score distributions, and
565 rater-wise agreement statistics. These diagnostics show that most disagreements are within one
566 ordinal level and that severe sign flips, e.g., human undersized vs. Gemini oversized, are rare.

567 .2 Quantitative Evaluation

568 Using LAION-Aes and Q-Align-IQ, we determine that the our pipeline does not degrade quality by
569 a significant ammount: on average, quality score decreases from 4.76 to 4.73, and Aesthetic score
570 decreases from 5.88 to 5.80. For context, [cite] consider any aesthetic score over 5 as sufficient
571 for synthetic training. We also show that images are semantically consistent through our pipeline,
572 according to a variety of embedding models [todo].

Table 16: **Human anchor selection in Task 2.** Products are paired with a human anchor according to their characteristic length, enabling interpretable product-to-human scale evaluation across small, medium, large, and oversized products.

Product characteristic length	Human anchor	Anchor length	#Entries
< 30 cm	human hand	19.3 cm	151
30–60 cm	human head/face	24.0 cm	77
60–100 cm	human foot/leg	26.5 cm	31
≥ 100 cm	full human body	170.0 cm	41

Table 17: **Task 3 construction protocol.** Task 3 is built from failed generations in Task 1 and Task 2, and each selected source image is converted into two correction prompts.

Component	Description
Source images	Failed or imperfect generations from Task 1 and Task 2. In the final benchmark, Task 3 uses 100 source images: 61 from T1 and 39 from T2.
Source models	T1 source images are generated by FLUX.2; T2 source images are generated by Qwen-Image.
Quality control	We remove images with severe synthesis failures, missing/unidentifiable target objects, duplicate main objects, or heavy occlusion/blur that would make scale correction ill-defined. Selection is not based on downstream editing success.
S4: Hard auto-discovery	The model is given the erroneous image and reference scale information, and must decide whether a correction is needed and which object to resize.
S5: Precise scale instruction	The model is directly given the object to resize and the exact scale factor, isolating scale-instruction following from error diagnosis.
Expansion	Each source image yields two prompts, one S4 and one S5, producing 200 Task 3 entries.

573 .3 Qualitative Evaluation

574 .4 Model Ablation Study

575 To examine the value of our pretraining, we conduct an ablation study where we vary the insertion
576 model in *Rescale*. Most importantly, we use the InsertAnything [44] DiT checkpoint, to show the
577 effects of our pretraining. Additionally, we try Creatilayout and Bifrost [27]. The results of this
578 investigation are shown in 22.

cite

579 .5 Pretraining Ablation Study

580 Results are shown in 24.

581 .6 Benchmark Release Format and Metadata

Table 18: **Task 3 visual quality control statistics.** We report the number of source candidates before and after QC, together with rejection reasons.

Source task	Candidates	Kept	Missing object	Duplicate object	Severe synthesis failure
T1	TBD	61	TBD	TBD	TBD
T2	TBD	39	TBD	TBD	TBD

Table 19: **Per-annotator agreement with the aggregate human consensus.** For each annotator, the reference is the aggregate human consensus constructed from all available raters using modal score aggregation with median-rounded tie breaking. Annotator identities are anonymized.

Annotator	Pairs	Exact (%) $\uparrow$	≤ 1 (%) $\uparrow$	MAE $\downarrow$	QWK $\uparrow$	r $\uparrow$	Mean diff.
Annotator A	506	70.95	95.26	0.3478	0.8087	0.8181	+0.0040
Annotator B	472	64.19	94.28	0.4280	0.7125	0.7183	-0.0593
Annotator C	519	65.51	94.99	0.4027	0.7853	0.7931	-0.0366
Annotator D	517	70.41	99.03	0.3056	0.8204	0.8404	-0.0387
Annotator E	522	61.88	90.42	0.4828	0.6006	0.6193	-0.0460
Annotator F	513	70.18	96.10	0.3411	0.7947	0.7955	+0.0331
Annotator G	511	55.19	90.02	0.5519	0.7433	0.7700	-0.0039
Annotator H	516	54.26	90.12	0.5562	0.6796	0.6813	+0.0252
Annotator I	526	73.76	96.01	0.3023	0.8371	0.8396	-0.0133
Mean $\pm$ std	511 $\pm$ 16	65.15 $\pm$ 6.98	94.03 $\pm$ 3.16	0.4132 $\pm$ 0.0989	0.7536 $\pm$ 0.0773	0.7640 $\pm$ 0.0760	–
Min–Max	472–526	54.26–73.76	90.02–99.03	0.3023–0.5562	0.6006–0.8371	0.6193–0.8404	–

Table 20: **Leave-one-rater-out human reliability.** For each annotator, the reference consensus is rebuilt from the remaining eight annotators only, using modal score aggregation with median-rounded tie breaking. This removes self-inclusion bias from the human reliability estimate. Annotator identities are anonymized and match Table 19.

Annotator	Ref. raters	Pairs	Exact (%) $\uparrow$	≤ 1 (%) $\uparrow$	MAE $\downarrow$	QWK $\uparrow$	r $\uparrow$
Annotator A	8	504	63.69	94.84	0.4246	0.7721	0.7862
Annotator B	8	515	59.61	93.79	0.4757	0.6783	0.6841
Annotator C	8	568	57.04	93.66	0.5000	0.7268	0.7374
Annotator D	8	558	64.70	98.39	0.3692	0.7737	0.7996
Annotator E	8	576	56.25	90.80	0.5347	0.5663	0.5885
Annotator F	8	553	60.76	95.48	0.4412	0.7430	0.7434
Annotator G	8	564	48.23	88.48	0.6365	0.6915	0.7271
Annotator H	8	565	47.26	90.27	0.6248	0.6397	0.6411
Annotator I	8	574	64.46	94.43	0.4146	0.7608	0.7667
Mean $\pm$ std	8	553 $\pm$ 26	58.00 $\pm$ 6.56	93.35 $\pm$ 3.03	0.4913 $\pm$ 0.0928	0.7058 $\pm$ 0.0695	0.7193 $\pm$ 0.0695
Min–Max	8	504–576	47.26–64.70	88.48–98.39	0.3692–0.6365	0.5663–0.7737	0.5885–0.7996

Table 21: **Gemini–human alignment with image-level bootstrap confidence intervals.** Gemini-3.1-Pro-preview is evaluated with five repeated samples per pair and majority/median aggregation. Confidence intervals are computed by resampling images rather than individual object pairs.

Split	Images / Pairs	Exact (%) $\uparrow$	≤ 1 (%) $\uparrow$	MAE $\downarrow$	r $\uparrow$	QWK $\uparrow$
Task 1: S1–S2	185 / 427	61.83 [56.69, 66.96]	96.49 [94.16, 98.35]	0.4169 [0.3565, 0.4786]	0.8473 [0.8108, 0.8774]	0.8373 [0.7968, 0.8696]
Task 2: S3	100 / 100	73.00 [64.00, 81.02]	100.00 [100.00, 100.00]	0.2700 [0.1898, 0.3600]	0.4701 [0.2340, 0.6517]	0.4677 [0.2278, 0.6459]
Task 1+2	285 / 527	63.95 [59.45, 68.45]	97.15 [95.29, 98.72]	0.3890 [0.3373, 0.4416]	0.8328 [0.7970, 0.8636]	0.8234 [0.7839, 0.8563]

Table 22: **OLD Results of Model Ablation Study**

Method	Text Alignment	Visual Consistency				Generation Quality	
Model	CLIP-T (%) $\uparrow$	CLIP-I (%) $\uparrow$	DINO (%) $\uparrow$	SSIM (%) $\uparrow$	SSIM-HF (%) $\uparrow$	LAION-Aes $\uparrow$	Q-Align-IQ $\uparrow$
Ours		89.7	74.8	41.2	71.8	3.83	3.10
InsertAnything		90.2	77.6	42.1	73.2	3.87	2.93
Bifrost		86.0	68.2	38.3	73.0	3.75	2.65
CreatiLayout		79.7	45.1	29.0	68.1	3.60	3.04

Table 23: **NEW Results of Model Ablation Study**

Method	Text Alignment	Visual Consistency				Generation Quality	
Model	CLIP-T (%) $\uparrow$	CLIP-I (%) $\uparrow$	DINO (%) $\uparrow$	SSIM (%) $\uparrow$	SSIM-HF (%) $\uparrow$	LAION-Aes $\uparrow$	Q-Align-IQ $\uparrow$
Ours		86.4	69.8	50.6	77.6	2.24	3.58
InsertAnything		87.7	74.7	49.4	76.9	2.30	3.61

Table 24: **Results of Pretrain Ablation Study**

Method	Text Alignment	Visual Consistency				Generation Quality	
Model	CLIP-T (%) $\uparrow$	CLIP-I (%) $\uparrow$	DINO (%) $\uparrow$	SSIM (%) $\uparrow$	SSIM-HF (%) $\uparrow$	LAION-Aes $\uparrow$	Q-Align-IQ $\uparrow$
baseline		74.2	30.6	44.1	75.8	3.19	1.91
baseline guidance30 depth dal		79.5	47.7	45.2	75.7	3.58	2.19
baseline guidance30 depth		79.4	47.1	45.0	75.6	3.58	2.19
baseline guidance30		77.6	44.8	40.8	74.2	3.40	2.13
baseline guidance30 depth dal hf		79.4	47.3	45.3	75.8	3.57	2.19

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 915 applicable), such as the institution conducting the review.

916 **16. Declaration of LLM usage**

917 Question: Does the paper describe the usage of LLMs if it is an important, original, or
 918 non-standard component of the core methods in this research? Note that if the LLM is used
 919 only for writing, editing, or formatting purposes and does *not* impact the core methodology,
 920 scientific rigor, or originality of the research, declaration is not required.

921 Answer: **[TODO]**
 922 Justification: **[TODO]**
 923 Guidelines:
 924 • The answer [N/A] means that the core method development in this research does not
 925 involve LLMs as any important, original, or non-standard components.
 926 • Please refer to our LLM policy in the NeurIPS handbook for what should or should not
 927 be described.